



1	2	3	4	5

6	7	8	9	10

11	12	13	14	15

16	17	18	19	20

1. The output of a logic gate can be one of two ____.

- a. Inputs
- b. Gates
- c. States
- d. none

2. Logic states can only be ____ or 0.

- a. 3
- b. 2
- c. 1
- d. 0

3. The output of a ____ gate is only 1 when all of its inputs are 1.

- a. NOR
- b. XOR
- c. AND
- d. NOT

4. A NAND gate is equivalent to an AND gate plus a gate put together.

- a. NOR
- b. NOT
- c. XOR
- d. none

5. Half adder circuit is ____.

- a. Half of an AND gate
- b. A circuit to add two bits together
- c. Half of a NAND gate
- d. none of above



6. Which logic gate does the truth table shown in figure 1 have?

- a. A two-input AND gate.
- b. A two-input OR gate.
- c. An exclusive OR gate.
- d. An exclusive NOR gate.

7. What Boolean expression describes the output X of the arrangement shown in figure 2?

- a. $X = A + B + C$
- b. $X = A.(B + C)$
- c. $X = (A.B) + C$
- d. $X = A + (B.C)$

8. In the Karnaugh map shown in figure 3, which of the loops shown represents a legal grouping?

- a. x
- b. y
- c. z
- d. w

9. Half adder consists of. andGates

- a. EX-OR, AND
- b. EX-OR, OR
- c. EX-OR, NOT
- d. None of the above

10. In half adder EX-OR gate output is

- a. Carry
- b. Remainder
- c. Sum
- d. None of the above

11. Full adder is constructed by using

- a. Two Half Adder & one OR gate
- b. Two OR gate & one HA
- c. One HA & two OR gate
- d. One OR gate & one HA

12. A half adder gives..... Output(s)

- a. 1
- b. 2
- c. 3
- d. none of the above

13. A full adder gives..... Output(s)

- a. 1
- b. 2
- c. 3
- d. non of this



14. The output of Half adder is in the form of.

- a. Sum
- b. carry
- c. sum & carry
- d. none of these

15. Logic 0 is represented by GND practically

- a. True
- b. False

16. $A'B' = (AB)'$

- a. True
- b. False

17. $AB = (A' + B)'$

- a. True
- b. False

18. The red light on the LED indicates to result = 0.

- a. True
- b. False

19. The output of an AND gate with three inputs, A, B, and C, is HIGH when

- a. $A = 1, B = 1, C = 0$
- b. $A = 0, B = 0, C = 0$
- c. $A = 1, B = 1, C = 1$
- d. $A = 1, B = 0, C = 1$

20. The Boolean expression for a 3-input AND gate is

- a. $X = AB$
- b. $X = ABC$
- c. $X = A + B + C$
- d. $X = AB + C$



Figures

<table><tr><th>A</th><th>B</th><th>C</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	C	0	0	0	0	1	1	1	0	1	1	1	1	<pre>graph LR; A --- AND1[AND]; B --- AND1; AND1 --- AND2[AND]; C_bar[NOT C] --- AND2; AND2 --- X</pre>
A	B	C														
0	0	0														
0	1	1														
1	0	1														
1	1	1														
Figure 1	Figure 2															

Figure 3	